Special Topics

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This manual contains a guide on how to structure a proposal for the **optional** special topics component of CMSC 131. If you choose to undertake a special topic, you are entitled to the **full substitution of your grade** in a regular component of your choice from the following: *Quizzes, Assignments, and Reportings*, *Midterm Exam*, or *Final Exam*. However, if you feel confident in your class standing, you may choose to forego a special topic and (safely) ignore this manual.

Overview

A special topic is defined as a specific application of concepts learned in the course. To that end, you are required to formulate a proposal for an **advanced project** that is related to CMSC 131. The project must be feasible in the span of **one month** and **its proposal must be approved by the instructor.** Said instructor has taken the liberty of jotting down possible topics for which you can propose a project. These topics include, but are not limited to:

1. Game Design and Development
2. Computer Security
3. Operating Systems
4. Embedded Systems
5. Tiny Machine Learning
6. Decompilation and Disassembly
7. Computer Architecture and Processor Design
8. Low-Level Optimization and Performance Tuning
9. Bootloaders and Firmware Development
10. Device Drivers and Hardware Interfaces
11. Memory Management and Allocation
12. Debugging Tools and Techniques
13. Compiler Design and Code Generation
14. Binary Analysis and Reverse Engineering
15. Real-Time Systems Programming
16. Microcontroller Programming

The Proposal

Your special topics proposal must contain the following sections:

1. Introduction
2. Objectives
3. Scope and Limitations
4. Methodology
5. Timeline
6. Expected Deliverables
7. Resources or References

The **Introduction** section must summarize the entirety of your proposed project. It must highlight the broader field where your project belongs, and it must also highlight how it is related to the course. The summary must also mention the expected deliverables of the project.

The **Objectives** section must enumerate the specific goals of your project. These should be clear, measurable, and achievable within the one-month timeframe.

The **Scope and Limitations** section must define the boundaries of your project. It should specify what will be included and, equally important, what will be excluded or simplified. This section should also acknowledge any technical constraints or challenges you anticipate.

The **Methodology** section must describe your approach to completing the project. This includes the programming languages, tools, libraries, and techniques you will use. It should also outline the major steps or phases of development.

The **Timeline** section must present a realistic schedule for completing your project. Break down the work into weekly milestones or phases, ensuring all tasks can be completed within one month.

The **Expected Deliverables** section must list the concrete outputs of your project. This typically includes source code, documentation, a demonstration or presentation, and any other materials that will be submitted for evaluation. Keep in mind that you will also be required to present your work before an audience by the end of the semester.

The **Resources or References** section must cite any technical resources, research papers, tutorials, libraries, or documentation that will guide your project. Use proper citation format (APA 7th Edition) for any academic references.